

# SOFDY: Physics-Based Soft Attachment Simulation in Motion to Enhance Realism of Inertial Data Synthesis

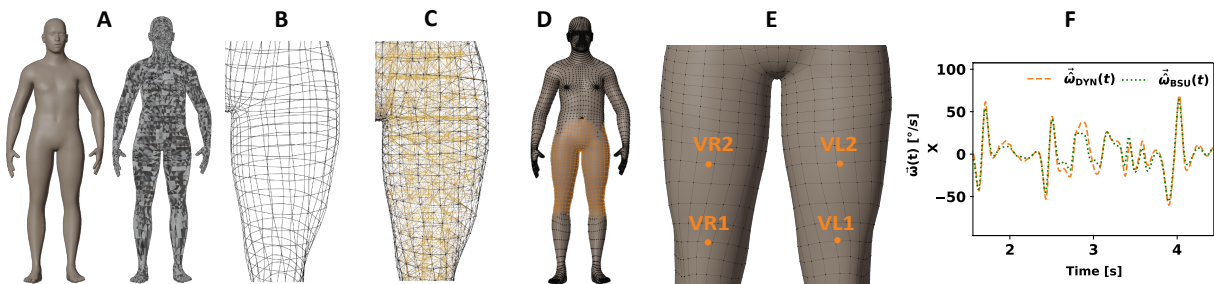
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**Figure 1: Overview of the soft attachment simulation methodology. A: The surface mesh of the reference BSU model is converted into a tetrahedral mesh. B: Mesh structure before tetrahedralisation. C: Mesh structure after tetrahedralisation; inner vertices highlighted in orange. D: Thigh area selected for soft attachment simulation. E: Selected vertices for synthesising acceleration and angular velocity. F: Error and correlation analysis.**

## Abstract

We present an approach to simulate and validate the effect of dynamic soft tissue and sensor attachment on inertial sensor data. Our approach extends sensor data synthesis from personalised biomechanical dynamics and human body surface models. Tetrahedralisation is employed to the surface mesh of body models to represent internal tissue dynamics. A position-based dynamics algorithm is employed to simulate soft attachment motion. We synthesised acceleration and angular velocity data for the thigh segment during jumping trials of 13 participants. We compare the synthesised inertial sensor data, with and without soft attachment simulation, to physical IMU measurements obtained concurrently to video motion capture data. Results show an average mean absolute error of  $19.98^\circ/s$  vs.  $21.45^\circ/s$  and  $2.19 m/s^2$  vs.  $2.25 m/s^2$ , with and without soft attachment simulation, respectively. The proposed soft attachment modelling approach showed lower error and higher correlation, along with considerably decreased error

standard deviation of up to 42% compared to the model without soft attachment. We conclude that dynamic soft attachment simulation can enhance existing body surface models and provide a more realistic representation of dynamic motion for the analysis and optimisation of wearable inertial sensor systems.

## CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing design and evaluation methods.

## Keywords

Soft sensor-body attachment, Multiscale modelling, Sensor data synthesis, Wearable inertial sensors

## ACM Reference Format:

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## 1 Introduction

Accurate simulation of dynamic motion related to soft tissue and sensor attachment is critical to enhance, e.g., wearable systems,



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human activity recognition (HAR) systems, and digital health platforms. Both soft tissue motion and the motion of wearable sensors, due to their attachments on the skin surface, can impact motion tracking quality but are often neglected in human body modelling and sensor synthesis approaches.

Current trend methods for IMU synthesis often rely on learned shape models and require large, resource-intensive datasets. In addition, learned shape models often come pretrained, without available code for retraining, mostly excluding soft attachment simulation. In addition, data-driven models approximate the visible surface deformations without considering underlying physical dynamics [16]. In current approaches from video and text prompts, e.g., [7, 8, 10, 11] joint positions are used to synthesise IMU sensor data, which does not reflect the actual sensor positioning and measurement conditions. Physics-based human body modelling approaches with soft attachment simulation have rarely been used in motion analysis. Therefore, the understanding of soft attachment dynamics is restricted and limits further model development.

In this work, we investigate the addition of a physics-based soft attachment simulation to enhance the realism of IMU data synthesis on an already published biomechanical kinematic and surface model (BSU) [19]. In particular, we provide the following contributions:

- (1) We present a framework to simulate soft attachment motion with a physics-based modelling approach to enhance the realism of IMU data synthesis. To accurately represent internal tissue and sensor attachment dynamics and prevent instability and collapse of the existing surface, we generate a tetrahedral mesh based on the surface mesh data from 13 study participants.
- (2) We evaluate the synthesis performance using Mean Absolute Error (MAE) and Pearson Correlation Coefficient (PCC) by comparing two models: (1) a model incorporating soft attachment dynamics (DYN) and (2) the baseline BSU model, assessed against physical IMU measurements. As a proof of concept, we analyse the thigh segment during jumping due to the expected substantial soft attachment displacement of the selected limb and the high-impact nature of jumping.

## 2 Related work

### 2.1 Data-driven deformation dynamics

Data-driven methods approximate soft attachment motion by learning patterns from (MoCap) data. Learned models often rely on computationally expensive 3D scans of deformations, but only a few explicitly capture soft attachment motion [16].

One of the earliest learned models was SCAPE [2], which introduced a framework for capturing human body shape variation across individuals and different poses. Building on SCAPE, the SMPL model was developed to provide a more efficient and realistic parametric representation of human bodies [12]. SMPL introduced a skinned, vertex-based structure that represents a wide variety of body shapes in different human poses. SMPL was later extended to provide pose-dependent soft tissue deformations to improve realism in motion-dependent shape changes. Expanding on SMPL, the Dyna model [16] was introduced to capture soft tissue dynamics by learning deformations from sequences of 3D scans.

Previous work approximated soft tissue deformations using learned statistical models rather than explicitly simulating soft attachment motion. While SMPL can be further used by the research community, the model training data and learning pipeline remain closed and prevent retraining and modification. In contrast, we employ a physics-based modelling approach to explicitly simulate soft attachment motion that provides physical interpretability and can be reproduced by the methods described in this work.

### 2.2 Physics-based deformation dynamics

Physics-based approaches simulate soft attachment dynamics using volumetric representations and force-based deformations.

One approach is the STMPL model [1], which extends SMPL [12] by a volumetric soft tissue layer [17]. Deformations were computed using 2D UV map representations of the 3D body shape, soft tissue, and external forces, before being mapped back into 3D. Another approach for soft tissue simulation was presented by the DRIPTA model [6]. DRIPTA employed a layered volumetric approach to combine a data-driven inner layer (VSMPL) with a finite element method (FEM)-based outer layer. The ADREDEEC model [9] introduced real-time dynamic effects to existing animations by attaching a viscoelastic soft tissue layer to the skeletal structure. The XPBD model [13] offers real-time simulation by operating on particle positions rather than forces. XPBD employs an implicit time-stepping method, adjusting particle positions in each step based on predefined volume and position constraints. In addition, damping mechanisms account for mechanical energy dissipation, which enhances realism in soft tissue dynamics.

The existing physics-based soft attachment simulation models offer various approaches to model tissue dynamics. However, the lack of available implementation details, code, training data, and dynamic restrictions limits the use for further research and development, except for the XPBD algorithm. In addition, unlike force-based FEM methods, XPBD operates on position constraints, making it more computationally efficient and easier to integrate while still providing physically plausible deformations [14].

## 3 Methods

### 3.1 Reference body model

As basis for this work, we used the combined, personalised biomechanical kinematics model and human surface (BSU) model [18]. BSU consists of a linked armature (bone-like structure) derived from MoCap markers and biomechanical equations, coupled with a surface mesh.

### 3.2 Soft attachment modelling

We employ the XPBD algorithm [13] to manipulate vertex positions in space, thus let meshes respond to external forces [14]. In a first step, the velocity in the direction of gravity is updated:  $\vec{v}_i' = \vec{v}_i + \Delta t * G$ , where  $i$  represents the vertex,  $\vec{v}_i$  the velocity in the direction of gravity,  $\Delta t$  the time step, and  $G$  the gravity parameter. The gravity parameter influences the vertex velocity by applying a downward force. Subsequently, the updated velocity is used to compute the new point position:  $\vec{p}_i' = \vec{p}_i + \Delta t * \vec{v}_i'$ , where  $\vec{p}_i$  is the current position of the vertex. Since the updated positions no longer strictly adhere

to the underlying model but are instead influenced by the velocity direction, edge constraints must be resolved to maintain structural consistency. The edge constraint ensures that the distances between neighbouring vertices remain consistent with the initial model following:

$$\Delta x_1 = -\frac{w_1}{w_1 + w_2 + \frac{\alpha}{\Delta t^2}} (l - l_0) \frac{\vec{x}_1 - \vec{x}_2}{|\vec{x}_1 - \vec{x}_2|} \text{ and} \quad (1)$$

$$\Delta x_2 = -\frac{w_2}{w_1 + w_2 + \frac{\alpha}{\Delta t^2}} (l - l_0) \frac{\vec{x}_2 - \vec{x}_1}{|\vec{x}_2 - \vec{x}_1|}, \quad (2)$$

where,  $w_i$  represents the weight of vertex  $i$ ,  $\alpha$  represents the compliance,  $\Delta t$  the time step,  $l$  the current vertex distance,  $l_0$  the distances computed during model initialisation, and  $\vec{x}_i$  the vertex position. The vertex weights influence the extent of the adjustments, determining to what extent each vertex is affected by the constraint.

Compliance (defined as the inverse of stiffness) controls how freely vertices move relative to each other. Once the vertex positions have been corrected, the velocities are updated another time by:  $\vec{v}_i' = \frac{\vec{p}_i - \vec{p}_i'}{\Delta t}$ , where  $\vec{v}_i'$  represents the updated vertex velocity,  $\vec{p}_i'$  the updated vertex position,  $\vec{p}_i$  the previous vertex position, and  $\Delta t$  the time step. Since the velocity depends on position updates, each vertex has an independent velocity. After the velocity update, the time step increments, and the XPBD loop proceeds to the next iteration.

### 3.3 Soft attachment implementation

For implementation, Blender [3] was selected based on compatibility with the original BSU model, avoidance of conversion issues (loss of armature or surface), and Blender's ability to edit vertex positions. In addition, we implemented a custom solver to control XPBD parameters during the simulation.

**Tetrahedralisation.** To simulate soft attachment dynamics using XPBD, the BSU surface mesh was supplemented by internal vertices through tetrahedralisation (see Fig. 1A). A volumetric mesh of tetrahedra distributes forces internally to provide an accurate simulation of elasticity and soft body dynamics. We used Delaunay triangulation [5]. Tetrahedralisation parameters were selected empirically: Resolution = 50, minQualityExp = -4, oneFacePerTet = True, and tetScale = 1.0.

**Vertex weights.** (1) In Blender, weight painting determines how much a bone influences a mesh vertex during linking armature and mesh [4]. A weight of 1 indicates full influence by the armature, while a weight of 0 denotes no influence. (2) In XPBD, the weight of individual vertices is used to solve the edge constraint to maintain structural consistency (see Sec. 3.2) and, therefore, drive soft tissue deformation. Consequently, we considered inner vertices as rigid and linked to the armature (Blender weights), while outer vertices were considered for soft attachment simulation (XPBD weights).

### 3.4 IMU data synthesis

To synthesise angular velocity and acceleration data, we used an additive acceleration model [18], which accounts for dynamic and gravitational acceleration, along with an angular velocity model derived from quaternion-based vertex orientation.

A total of four different vertices on the thighs were considered in this work. Two vertices were selected where the physical IMU sensors were located in the measurements (see Fig. 1E, VL1, VR1). The

other two vertices were located near the midpoint of the thighs (see Fig. 1E, VL2, VR2), where soft tissue deformation and muscle interaction are most pronounced.

### 3.5 XPBD parameter optimisation

To maximise simulation stability and physical realism, we systematically evaluated 32 XPBD parameter combinations, including compliance (0.0 and  $1 \times 10^{-5}$ ), weight (0.1, 0.5, 1.0, and 5.0), and gravity (0.01, 0.1, 1.0, and 10.0). Optimisation for  $\theta^*$  was performed by minimising MAE and maximising PCC:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} [\operatorname{MAE}(x, \hat{x}) - \operatorname{PCC}(x, \hat{x})], \quad (3)$$

where  $\theta = \{\text{compliance, gravity, weight}\}$  denotes the set of optimised parameters,  $x$  the measured inertial sensor time series, and  $\hat{x}$  the simulated (DYN) sensor time series.

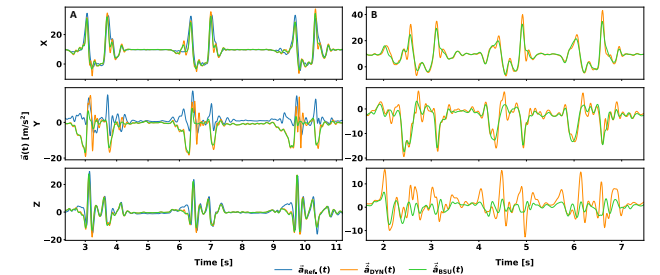
### 3.6 Evaluation data

We analysed 13 participants (7 males, 6 females; age:  $26.3 \pm 6.4$  years; height:  $1.74 \pm 0.10$  m; weight:  $63.1 \pm 9.5$  kg) during a jumping activity, with each participant performing three countermovement jumps. Marker setup and IMU placements are shown in [18]. All volunteers provided written informed consent prior to participation. Ethics approval was granted by the ethics committee of Friedrich-Alexander University Erlangen-Nuremberg (345\_19 B).

### 3.7 Evaluation metrics

Synthesis performance was assessed by MAE between the simulated (DYN and BSU) sensor time series and the measured inertial sensor time series for each sensor modality, virtual sensor instance at the body area of interest, and participant. In addition, PCC was derived to assess the linear relationship between simulated (DYN and BSU) and measured inertial sensor data.

## 4 Results

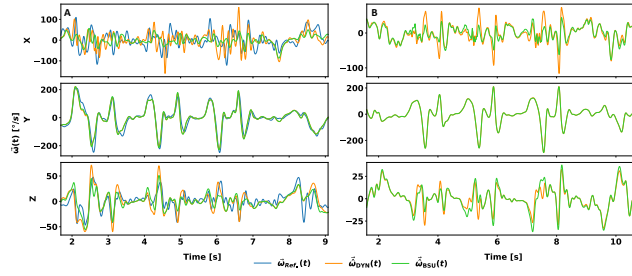


**Figure 2: Excerpt of measured ( $\vec{a}(t)$ ) and simulated acceleration time series for BSU ( $\vec{a}_{BSU}(t)$ ) and DYN ( $\vec{a}_{DYN}(t)$ ) during jumping. A: VL1 of Participant P10. B: VL2 of Participant P2 without a reference signal due to the absence of a physical IMU attached at the midpoint of the thigh.**

Figure 2A shows an excerpt of simulated acceleration signals for BSU ( $\vec{a}_{BSU}(t)$ ) and DYN ( $\vec{a}_{DYN}(t)$ ) models, compared to measured IMU sensor time series ( $\vec{a}_{Ref}(t)$ ) at sensor position VL1 (above knee). The DYN model shows higher positive and negative peaks and greater variability in the angular velocity time series compared to BSU. Figure 2B shows greater signal deviations between BSU

and DYN at sensor position VL2 (thigh midpoint). In particular, on the axes that do not align with the main axis of movement (y and z axes), additional signal variations can be detected.

Figure 3A shows an excerpt of simulated angular velocity time series of BSU without soft attachment ( $\vec{\omega}_{BSU}(t)$ ) and our DYN model with soft attachment ( $\vec{\omega}_{SK}(t)$ ), compared to measured IMU sensor time series ( $\vec{\omega}(t)$ ). Similar to the acceleration, the DYN model exhibits higher positive and negative peaks and increased signal variance on the x and z axes in the angular velocity time series (main axis of rotation around y in the sagittal plane). Figure 3B shows greater signal deviations when BSU is compared to DYN at sensor position VR2 (thigh midpoint); however, less pronounced for angular velocity than for acceleration.



**Figure 3:** Excerpt of measured ( $\vec{\omega}(t)$ ) and simulated angular velocity time series for BSU ( $\vec{\omega}_{BSU}(t)$ ) and DYN ( $\vec{\omega}_{DYN}(t)$ ) during jumping. A: VL1 of Participant P2. B: VR2 of Participant P2 without a reference signal due to the absence of a physical IMU attached at the midpoint of the thigh.

**Table 1:** Comparison of model performance for angular velocity and acceleration data for vertices averaged across VL1 and VR1 per participant. BSU model without and DYN model with soft attachment simulation using the XPBD algorithm after parameter optimisation. MAE, PCC, and SD are shown for each model.

|             | BSU                      |        |                  |        | DYN                      |        |                  |        |
|-------------|--------------------------|--------|------------------|--------|--------------------------|--------|------------------|--------|
|             | Acc. (m/s <sup>2</sup> ) |        | Angl. vel. (°/s) |        | Acc. (m/s <sup>2</sup> ) |        | Angl. vel. (°/s) |        |
| Participant | MAE                      | PCC    | MAE              | PCC    | MAE                      | PCC    | MAE              | PCC    |
| P1          | 2.35                     | 0.45   | 17.29            | 0.76   | 1.94                     | 0.64   | 17.04            | 0.74   |
| P2          | 2.80                     | 0.37   | 20.46            | 0.62   | 2.21                     | 0.64   | 17.43            | 0.66   |
| P3          | 2.66                     | 0.42   | 23.45            | 0.49   | 2.64                     | 0.62   | 23.27            | 0.49   |
| P4          | 2.21                     | 0.50   | 17.93            | 0.58   | 1.79                     | 0.68   | 17.57            | 0.53   |
| P5          | 2.01                     | 0.48   | 15.36            | 0.52   | 1.82                     | 0.60   | 19.07            | 0.57   |
| P6          | 2.36                     | 0.49   | 17.53            | 0.70   | 1.81                     | 0.69   | 15.55            | 0.72   |
| P7          | 2.50                     | 0.43   | 23.21            | 0.56   | 2.20                     | 0.65   | 18.82            | 0.61   |
| P8          | 1.26                     | 0.82   | 24.25            | 0.53   | 2.11                     | 0.49   | 20.70            | 0.56   |
| P9          | 2.09                     | 0.54   | 17.08            | 0.68   | 2.44                     | 0.58   | 17.91            | 0.67   |
| P10         | 2.49                     | 0.45   | 21.33            | 0.68   | 2.40                     | 0.64   | 22.30            | 0.76   |
| P11         | 2.27                     | 0.58   | 20.39            | 0.64   | 2.44                     | 0.55   | 19.97            | 0.58   |
| P12         | 2.38                     | 0.71   | 40.85            | 0.40   | 2.70                     | 0.52   | 33.40            | 0.58   |
| P13         | 1.89                     | 0.54   | 19.73            | 0.58   | 1.97                     | 0.49   | 16.77            | 0.61   |
| Mean        | 2.25                     | 0.52   | 21.45            | 0.60   | 2.19                     | 0.60   | 19.98            | 0.62   |
| ± SD        | ± 0.37                   | ± 0.12 | ± 6.19           | ± 0.09 | ± 0.30                   | ± 0.07 | ± 4.42           | ± 0.08 |

Tab 1 shows the optimised performance after adding the soft attachment simulation to the synthesis framework (DYN). The DYN model demonstrated improved performance compared to the BSU model. Regarding acceleration, the mean MAE for the DYN model was  $2.19\text{m/s}^2$ , slightly lower than the  $2.25\text{m/s}^2$  for the BSU model. The DYN model also exhibited a lower SD of  $0.30\text{m/s}^2$  compared to  $0.37\text{m/s}^2$  for the BSU model, suggesting improved performance

consistency. For angular velocity, the mean MAE for the DYN model was  $19.98^\circ/\text{s}$ , a decrease from  $21.45^\circ/\text{s}$  for the BSU model. In addition, the DYN model showed a lower standard deviation (SD) of  $4.42^\circ$  compared to  $6.19^\circ$  for the BSU model, indicating greater reliability across participants.

## 5 Discussion and Conclusion

Figures 2A and 3A demonstrate higher and lower signal peaks in the DYN model compared to the BSU model. The discrepancy results from the incorporation of an additional layer of complexity and motion to the BSU model's surface. The IMU positions were chosen according to the manufacturer's recommendations [15], thus, the positions were closer to the joints to avoid excessive adipose tissue and minimise soft attachment artefacts, which may explain the moderate MAE improvements observed in this work. However, other applications using alternative IMU placements on soft attachment, e.g., IMU on adipose tissue or a smartphone in a trouser pocket/waist bag, are expected to benefit more from the soft attachment simulation approach. The effect of soft attachment can be clearly observed in Fig. 2B and Fig. 3B. Both data examples show more pronounced signal dynamics for the DYN model compared to the BSU model at mid-thigh sensor positions (VR2 and VL2), as opposed to sensor positions at the distal end near the knee (VL1 and VR1 in Fig. 2a and Fig. 3A). The results align with the common expectation that the greater proportion of soft tissue in the central thigh area leads to more pronounced dynamics in the DYN model.

The overall MAE improvement by the DYN model, compared to the BSU model, was  $\sim 6.9\%$  for angular velocity and  $\sim 2.7\%$  for acceleration (see Tab. 1). PCC decreased slightly by approx.  $8.3\%$  across sensor modalities. While differences in MAE and PCC are minor, error variance (i.e. SD) across study participants decreased substantially ( $\sim 18.9\%$  for acceleration and  $\sim 28.6\%$  for angular velocity). In addition, PCC SD decreased by  $\sim 41.7\%$  for acceleration and by  $\sim 11.1\%$  for angular velocity. We attribute the SD reduction compared to BSU to a more reliable and robust DYN performance.

To minimise the observed error between synthesised and measured data, a parameter optimisation was conducted. In the future, characterisation of the XPBD parameters should be performed to improve practical precision, e.g. by calibrating the gravity parameter to real acceleration values.

In this work, as proof of concept, only the thighs during jumping trials were simulated with soft attachment dynamics (see Fig. 1D), which restricts the generalisation of results. The impact of full-body surface simulation on the overall synthesis results remains uncertain and warrants further investigation.

We implemented the physics-based XPBD approach in Blender to enhance the existing BSU model with a tetrahedral mesh and optimised the XPBD parameters (gravity, compliance, weight). Synthesis performance was assessed visually and quantitatively. We showed that DYN follows the signal patterns of BSU, with additional dynamics. Compared to BSU, the DYN model exhibited a lower MAE, higher PCC, and generally lower SDs of approx.  $25\%$ . We believe that this work contributes a physics-inspired method to enhance the realism of IMU data synthesis. Future research is warranted that extends the simulation to the whole body, increases the number of study participants, and investigates a wider range of physical activities.

## References

- [1] Anton Agafonov and Lihi Zelnik-Manor. 2024. STMPL: Human Soft-Tissue Simulation. arXiv:2403.08344
- [2] Dragomir Anguelov, Praveen Srinivasan, Daphne Koller, Sebastian Thrun, Jim Rodgers, and James Davis. 2005. SCAPE: shape completion and animation of people. (July 2005), 408–416. doi:10.1145/1186822.1073207
- [3] Blender Foundation. 2025. Blender. <https://www.blender.org/>. [Accessed: 09-Sep-2024].
- [4] Blender Foundation. 2025. Blender 4.3 Manual, Weight Painting Introduction. [https://docs.blender.org/manual/en/latest/sculpt\\_paint/weight\\_paint/introduction.html#the-weighting-color-code](https://docs.blender.org/manual/en/latest/sculpt_paint/weight_paint/introduction.html#the-weighting-color-code) [Accessed: 23.02.2025].
- [5] Mark de Berg, Otfried Cheong, Marc van Kreveld, and Mark Overmars. 2019. *Computational Geometry: Algorithms and Applications*. Springer, Berlin, Germany.
- [6] Meekyoung Kim, Gerard Pons-Moll, Sergi Pujades, Seungbae Bang, Jinwook Kim, Michael J. Black, and Sung-Hee Lee. 2017. Data-driven physics for human soft tissue animation. *ACM Transactions on Graphics* 36, 4 (Aug. 2017), 1–12. doi:10.1145/3072959.3073685
- [7] Hyeokhyen Kwon, Catherine Tong, Harish Haresamudram, Yan Gao, Gregory D. Abowd, Nicholas D. Lane, and Thomas Plötz. 2020. IMUTube: Automatic Extraction of Virtual on-body Accelerometry from Video for Human Activity Recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 3 (Sept. 2020), 1–29. doi:10.1145/3411841
- [8] Hyeokhyen Kwon, Bingyao Wang, Gregory D. Abowd, and Thomas Plötz. 2021. Approaching the Real-World: Supporting Activity Recognition Training with Virtual IMU Data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 3 (Sept. 2021), 1–32. doi:10.1145/3478096
- [9] Caroline Larboulette, Marie-Paule Cani, and Bruno Arnaldi. 2005. Dynamic skinning: adding real-time dynamic effects to an existing character animation. (May 2005), 87–93. doi:10.1145/1090122.1090138
- [10] Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, and Thomas Plötz. 2024. IMUGPT 2.0: Language-Based Cross Modality Transfer for Sensor-Based Human Activity Recognition. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 3 (Sept. 2024), 112:1–112:32. doi:10.1145/3678545
- [11] Zikang Leng, Hyeokhyen Kwon, and Thomas Ploetz. 2023. Generating Virtual On-body Accelerometer Data from Virtual Textual Descriptions for Human Activity Recognition. In *Proceedings of the 2023 ACM International Symposium on Wearable Computers (ISWC '23)*. Association for Computing Machinery, New York, NY, USA, 39–43. doi:10.1145/3594738.3611361
- [12] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. 2015. SMPL: a skinned multi-person linear model. *ACM Trans. Graph.* 34, 6 (Oct. 2015), 248:1–248:16. doi:10.1145/2816795.2818013
- [13] Miles Macklin, Matthias Müller, and Nuttapong Chentanez. 2016. XPBD: position-based simulation of compliant constrained dynamics. (Oct. 2016), 49–54. doi:10.1145/2994258.2994272
- [14] Mathias Müller. 2025. XPBD, Extended Position Based Dynamics. <https://matthias-research.github.io/pages/tenMinutePhysics/09-xpbd.pdf> [Accessed: 26.02.2025].
- [15] Noraxon. 20.08.24. myoMOTION Research PRO & Clinical Hardware User Manual. <https://www.noraxon.com/download/myomotion-system-user-manual/?wpdmdl=18194&refresh=6736368f9fe1d1731606159>.
- [16] Gerard Pons-Moll, Javier Romero, Naureen Mahmood, and Michael J. Black. 2015. Dyna: a model of dynamic human shape in motion. *ACM Transactions on Graphics* 34, 4 (July 2015), 1–14. doi:10.1145/2766993
- [17] Hang Si. 2015. TetGen, a Delaunay-Based Quality Tetrahedral Mesh Generator. *ACM Trans. Math. Software* 41, 2 (Feb. 2015), 1–36. doi:10.1145/2629697
- [18] Lena Uhlenberg and Oliver Amft. 2022. Comparison of Surface Models and Skeletal Models for Inertial Sensor Data Synthesis. In *2022 IEEE-EMBS International Conference on Wearable and Implantable Body Sensor Networks (BSN)*. IEEE. doi:10.1109/BSN56160.2022.9928504
- [19] Lena Uhlenberg, Adrian Derungs, and Oliver Amft. 2023. Co-simulation of human digital twins and wearable inertial sensors to analyse gait event estimation. *Frontiers in Bioengineering and Biotechnology* 11 (2023). <https://www.frontiersin.org/articles/10.3389/fbioe.2023.1104000>